

This is the Sedimentologica L^AT_EX template example and build test

Abstract

This is a test abstract used for evaluating the build of an article using the *Sedimentologica* L^AT_EX template using LuaL^AT_EX. It contains no meaningful scientific content, but contains useful guidelines for writing your article using the template.

If your author and affiliation lists and CRediT roles are long, the abstract may run over the first page in 'preprint' mode. If this is the case, please attempt to shorten your abstract; however, if it is not reasonably possible please contact support@sedimentologica.org. Use `\documentclass[preprint]{sedimentologica}` to test the length of your abstract.

Plain language summary

This file tests the *Sedimentologica* article template build using LuaL^AT_EX. A plain language summary is mandatory.

Translations

We encourage authors to provide a translation of your abstract and plain language summary into a language of your choice. If you wish to provide multiple translations, please contact the editorial team for guidance

If your translation language requires special fonts (e.g., Arabic), instructions are available in the template file and the README.md file. Should you need further help with this, please contact the editorial team admin@sedimentologica.org.

1 Introduction

As of 2026, *Sedimentologica* has moved to L^AT_EX to typeset accepted publications in the journal. The primary motivations for this switch are to:

- use an open-source, free software for typesetting,
- provide authors with a streamlined pre-print to publication workflow,
- lower the time investment required by our volunteer typesetting team, and,
- ensure all papers published in *Sedimentologica* have a unified design.

While the primary uses of the template are for typesetting by the *Sedimentologica* team and authors preparing their article for submission to *Sedimentologica* it also provides you with an easy way to generate a pre-print version that you can post on a pre-print server (e.g., [EarthArXiv](#)). Simply use `\documentclass[preprint]{sedimentologica}` in the preamble above.

Like the policy of *Seismica*, we do not require authors to use this template at initial submission. **However**, **minimum formatting requirements** apply and upon acceptance authors using L^AT_EX should submit their final version using the *Sedimentologica* L^AT_EX template. Not using the template may incur delays in copy-editing, proofing, and publication. For authors using Microsoft Word or LibreOffice etc, there may be greater delays incurred depending on the complexity of the document.

2 Minimum formatting and layout requirement

While *Sedimentologica* strongly encourages the use of our templates for articles submitted to the journal, you may submit your article without using the templates if the following conditions are met:

- Your article is written in a single dialect of English
- Your manuscript is a single PDF, DOC, DOCX, or ODT file containing all figures and in-text tables.
- Your data is made accessible via a repository such as Zenodo.
- You have added to your manuscript the following statements, after the conclusions and before the references
- Data and code availability
- Conflict of interest statement
- Artificial intelligence use statement
- Funding declaration

3 Language

All submitted documents must be in English. Authors are required to use a single variety of English consistently throughout the manuscript, such as Australian, British, Canadian, Commonwealth, or US English.

While *Sedimentologica* requires manuscripts to be written in English, we encourage authors to provide an abstract and plain language summary translation in their preferred language of choice.

Submitted documents should be thoroughly proof-read prior to submission to ensure they are free from grammatical and spelling errors.

Abbreviations that are not standard to the field are not to be used in the abstract and should be introduced in the corpus of the text. The use of standard abbreviations such as “Fm.” or “Mbr” respectively, for “Formation” or “Member” should be reserved for figures and tables, while full-length words are used in the corpus of the manuscript.

All units must be from the [International System of Units](#) or derivative thereof (i.e. metres rather than feet or kPa rather than PSI). Customary/non-SI units may be included in parentheses.

Utilise the commands and options available in [siunitx](#) for any quantity or number.

- `\ang[options]{angle}`
- `\num[options]{number}`
- `\unit[options]{unit}`
- `\qty[options]{number}{unit}`
- `\numlist[options]{numbers}`
- `\numproduct[options]{numbers}`
- `\numrange[options]{numbers}{number2}`
- `\qtylist[options]{numbers}{unit}`
- `\qtyproduct[options]{numbers}{unit}`
- `\qtyrange[options]{number1}{number2}{unit}`

3.1 Inclusive Language

[Sedimentologica](#) strongly encourages authors and co-authors to use inclusive and non-discriminatory language in their manuscripts. The American Psychological Association (APA) has produced a guide you may find useful for this purpose, which you can obtain from [apa.org](#).

Scientific research is always a team effort. Therefore, we encourage using “we” about the authors as much as using passive voice or third person.

77 *Sedimentologica* encourages authors to use both traditional place names and their English equivalents, where
78 appropriate, to acknowledge the cultural significance and diversity of the regions being described, e.g., "the
79 Kaikōura Canyon is located east of Te Wāhipounamu (the South Island of Aotearoa/New Zealand)".
80 When applicable, acknowledgement of traditional custodians and their lands should be added.

81 3.2 Politically Sensitive Geographic Names

82 As Earth Scientists, we often use geospatial data. In some cases the geographic name of a location may be
83 disputed, or the border of a territory has not been honoured or is disputed.

84 For cases of geographic name disputes (e.g. Gulf of Mexico/Gulf of America), authors should first determine if an
85 arbitration decision has been made, for example under the UNCLOS, and use that name. If no such decision has
86 been made, authors should use the name which is most recognised internationally, refer to appropriate United
87 Nations maps (e.g. [UN Map of the World](#)), or provide both names. The intent is to ensure the name used is as
88 neutral as possible.

89 3.3 Geologic Time

90 For geologic time, please use nomenclature of the International Chronostratigraphic Chart, the latest version can
91 be found at [text](#).

92 4 Figures and tables

93 This section outlines the general guidelines for figures and tables. Please refer to *Sedimentologica's* [figure and](#)
94 [table guidelines](#) for further detail.

95 The final document will have a two-column page layout. You can test the appearance of your tables and figures in
96 this layout by using the preprint option for the sedimentologica document class.

97 4.1 Dimensions

98 It is best practice to design and test your figures and tables at their intended final size. Figures and tables should
99 preferably be designed to fit within either a single column (e.g., [Figure 1](#)) or across two columns (e.g., [Figure 2](#))
100 in portrait page mode. However, landscape figures and tables are allowable, but only use these if absolutely
101 required for legibility. [Table 1](#) summarises the maximum dimensions allowable for figures and tables that fit
102 within a single page.

Intended Width	Maximum Width	Maximum Height
one column	86 mm	230 mm
two column	180 mm	230 mm
landscape	257 mm	160 mm

Table 1 – Summary of standard dimensions for figures and tables.

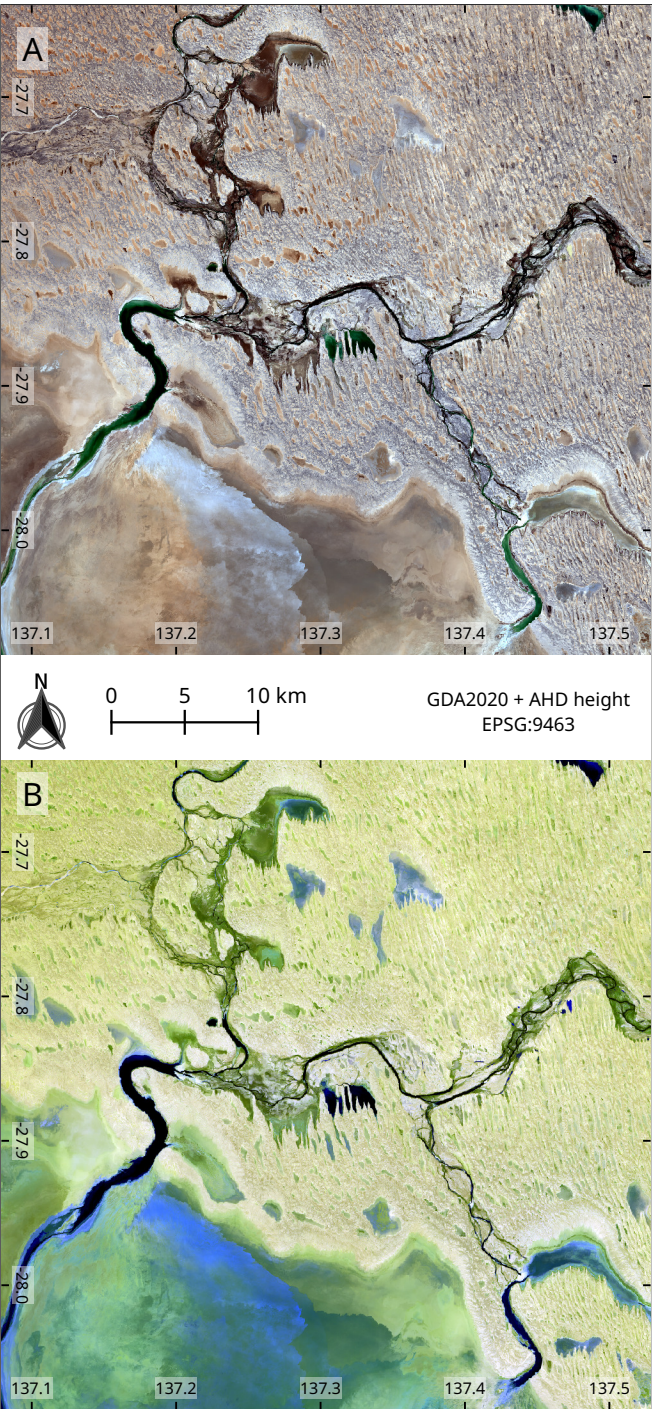


Figure 1 – Sentinel 2 L2A imagery (2025-10-18) of north-eastern Kati Thanda-Lake Eyre at the Warburton River discharge. **(A)** True colour image mapping bands 4, 3, and 2 to R, G, and B channels. **(B)** False colour image mapping bands 12, 11, and 2 to R, G, and B channels. Contains modified Copernicus Sentinel data [2025].

103 If a figure is being placed at the end of the document rather than near its insertion point define the figure optional
104 position argument using a ! or use \clearpage directly after the figure environment.

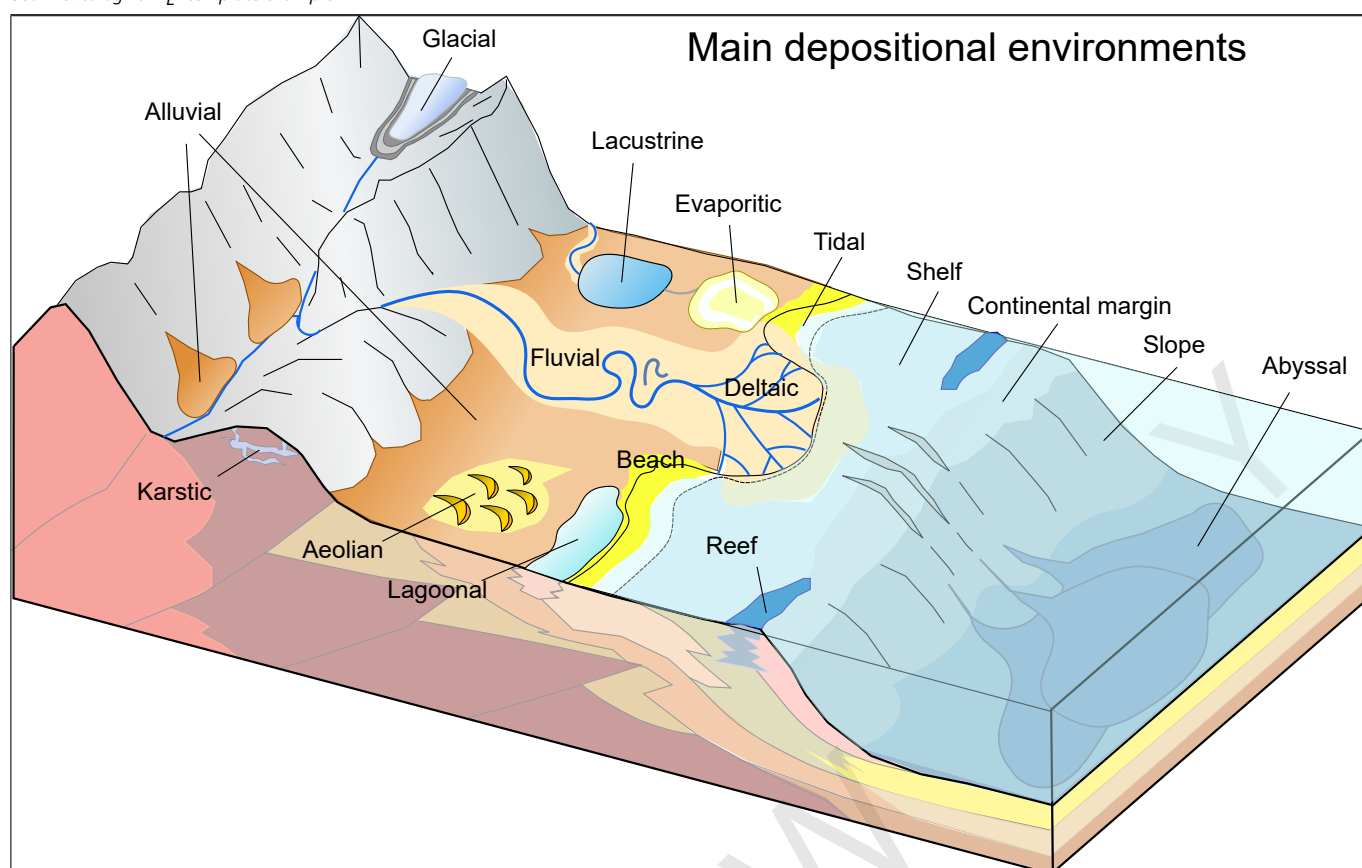


Figure 2 – Schematic diagram of the main depositional environments in sedimentary systems. Source: Mikenorton (2018)

4.2 Figure design

Figures must be designed with accessibility in mind. Do not rely on colour alone, use symbols or patterns to assist with differentiation on graphs and other diagrams.

Preferably use the [Noto Sans](#) font family as that is what we use for final production. However, any standard, sans-serif font (e.g., Arial, Helvetica, Verdana) can be used. Be consistent and only use one font family.

Use at most three (preferably no more than two) different font sizes for annotations on figures. The standard font size should be 10 pt Noto Sans or equivalent. Regardless of font choice, the minimum character height must be no less than 4.8 mm at final production size for accessibility purposes. Aim to use larger font sizes (e.g., 10 pt Noto Sans) to improve accessibility.

Colour schemes **must** be colour-vision deficiency safe (e.g. [Scientific Colour Maps](#)).

4.3 Maps

Map figures must have a scale, grid markers of some form, and state the coordinate reference system of their projection. [Figure 1](#) is an exemplar of minimum requirements for map figures.

Due to the nature of geology being an international science, some maps may have areas of disputed borders or regions. In these cases, please use the internationally recognised border or region where possible.

4.4 Indexing and cross-referencing

Figures and tables should be numbered in order of appearance and cross-referenced in text. In L^AT_EX, use `\autoref{ref}` to handle this. For example, these are cross-references to [Figure 2](#) and [Table 1](#) using `\autoref`.

For figures with sub-panels, add uppercase letters for indexing purposes as shown in [Figure 1A](#) and [Figure 1B](#). If additional sub-indices are required, use parenthesised lowercase Roman numerals, e.g., A(i).

4.5 Table formatting

Complex tables present a significant challenge in L^AT_EX for our volunteer production team. Please only use tables where needed, and keep them simple. Complex tables will incur production delays.

Use a table only if there isn't a simpler way to present your content, such as a paragraph of text or figure. Each table must be independently understandable, with a clear title or caption, defined abbreviations, and sufficient explanatory notes where needed. Tables should be numbered sequentially in the order they are cited in the manuscript. Design tables for readability and accessibility. The minimum font size should be Noto Sans 8 pt (or equivalent). Keep tables simple:

- use simple structure, clear column headings, consistent terminology, and SI units;
- avoid excessive decimal places, duplicated information, empty cells without explanation, and unnecessary formatting;
- do not use colour alone to communicate meaning.

Large or highly detailed datasets should be deposited in an appropriate repository and cited in the data availability statement rather than included as oversized manuscript tables. Datasets provided in a form such as CSV, ODS, XLSX, RData, JLD2, HDF5 etc. are better for study reproducibility.

Tables rules are formatted using [booktabs](#). This means you can use the `\toprule`, `\midrule`, and `\bottomrule` commands within the `tabular` environment.

The [nicematrix](#) package is also loaded, providing customisation for blocks and the X for the flexible column widths.

144 **Table 1** is a simple example of a one-column wide table with correct styling and using the NiceTabular*
 145 environment.

146 To ensure correct widths of tables, please use the provided custom commands

147 • `\onecolwidth` for single-column width

148 • `\twocolwidth` for two-column width

149 as the size of your NiceTabular* or tabular* environment.

150 For tables that will extend beyond a single page, longtable is available. If the longtable needs to be two
 151 columns wide, or in landscape orientation, please see the example_longtable.tex for an example of how to
 152 format these. We have placed the longtable at the end of this build to not clutter the primary information.

153 To place a table that is only a single column wide (i.e. small tables), use the table environment.

154 Otherwise, use the table* environment to place a table that will span both page columns.

155 *NOTE:* nicematrix and longtable will not work in the same table.

156 5 Mathematics

157 In general follow the [Short Math Guide \(PDF\)](#).

158 Use `\vec{x}` for vectors and `\mathbf{M}` for matrices, e.g.

159 `\vec{y} = \mathbf{X}\vec{\beta} + \epsilon`

160 will print:

$$\vec{y} = \mathbf{X}\vec{\beta} + \epsilon \quad (1)$$

161 In-line equations will look like this: $x^2 + y^2 = z^2$, while basic display equations will look like the following:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{i=n} x_j = \frac{x_1 + x_2 \dots + x_n}{n} \quad (2)$$

162 Multiline equations will look as follows:

$$\hat{\rho}_{\partial\rho}^2(\rho^2) = 1 - \frac{n-3}{n-(k+1)-1} \cdot (1-\rho^2) {}_2F_1\left(1, 1; \frac{n-(k+1)-1}{2}; 1-\rho^2\right) \quad (3)$$

163 Finally, aligned equations will look like this:

$$\sum (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (4)$$

$$\sum x_i (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (5)$$

$$\sum x_i^2 (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (6)$$

$$\sum x_i^3 (x_i - \varepsilon_1)(x_i - \varepsilon_2)(x_i - \varepsilon_3)(x_i - \varepsilon_4) = 0 \quad (7)$$

164 6 Chemical equations

165 The `mhchem` package is enabled in the template. Use `\ce{}` to wrap your chemistry. For example, `\ce{^{87}_{37}Rb+}`
 166 prints $^{87}_{37}\text{Rb}^+$.

167 7 Referencing

168 Appropriate citation of work is mandatory. While peer-reviewed literature should be the primary sources on which
 169 you rely, material considered to be *grey literature* such as pre-prints, theses, government reports, conference
 170 papers etc. should be appropriately cited.

171 The *Sedimentologica* class uses the `citation-style-languauge` package for referencing.

172 Use `\cite{}` for parenthetical citations, e.g., (Strata, 2065), and `\textcite{}` for in-line text citations, e.g., Helios
 173 et al. (2072) stated that this is an awesome template. Prefixes, suffices, and pointers can be added as optional
 174 arguments, e.g., `\cite[prefix = {e.g., }, suffix={and references therein}] {ExampleThesis}`
 175 will produce (e.g., Diamond, 2060 and references therein).

176 The following is to enable generation of a reference list with examples for an article (Helios et al., 2072), book (Strata,
 177 2065), book section (Core, 2068), dataset (Cryo & Ice, 2075), map (Carto, 2069), preprint (Plume & Geyser, 2074),
 178 report (Agency for Outer Worlds Geology, 2070), software (OrbitLab, 2073) and thesis (Diamond, 2060).

179 8 Conclusions

180 This is a mandatory section, replace this text.

181 Thanks for using the *Sedimentologica* L^AT_EX template for preparing your article.

182 Acknowledgements

183 Replace this text with your acknowledgements.

184 Data and code availability

185 Data used in this paper is hosted on Zenodo at doi.org/10.1000/182, \cite{ExampleCitation}.

186 The code used in this paper is available via [and](#) archived on Zenodo at doi.org/10.1000/182, \cite{ExampleCitation}.

187

188 Supplementary materials

189 If you have any additional supplementary material hosted in a repository for this manuscript please replace this with a state-

190 ment linking to the doi, and add the citation, e.g. Supplementary material for this paper can be found at doi.org/10.1000/182

191 \cite{ExampleCitation}.

192 Conflict of interest statement

193 The authors declare they do not have any tangible or perceived conflicts of interest with regard to this research.

194 Artificial intelligence use statement

195 All research, analysis, writing, and editing were performed without generative AI assistance.

196 Funding

197 Declare sources of funding for this research.

198 References

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Table 2 – Summary of laser conditions and materials analysed in each session.

Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence (J cm^{-2})	Measured Flu- ence (J cm^{-2})	Ablation Time (s)
2020-02-24	zircon	29	5	2		30
2020-02-24	glass	29	5	3.5		30
2020-02-24	glass	51	5	3.5		30
2020-02-26	zircon	29	5	2		30
2020-02-26	glass	29	5	3.5		30
2020-02-26	glass	51	5	3.5		30
2020-05-06	zircon	19	5	2		40
2020-05-06	glass	19	5	3.5		40
2020-05-06	glass	51	5	3.5		40
2020-05-08	glass	43	5	3.5		40
2020-05-11	zircon	29	5	2		40
2020-05-11	glass	29	5	3.5		40
2020-05-11	glass	51	5	3.5		40
2021-03-30	zircon	29	5	2		30
2021-03-30	glass	43	5	3.5		30
2021-03-31	zircon	30	5	2		40
2021-03-31	glass	43	5	3.5		40
2021-05-06	zircon	30	5	2		40
2021-05-06	glass	43	5	3.5		40
2021-09-06	zircon	30	5	2		30
2021-09-06	glass	43	5	3.5		30
2021-09-06	monazite	20	5	2		30
2022-01-19	baddeleyite	20	5	2		30
2022-01-19	zircon	20	5	2		30
2022-01-19	glass	43	5	3.5		30
2022-01-19	apatite	30	5	3.5		30
2022-02-01	monazite	13	5	2	1.9	30
2022-02-01	xenotime	13	5	2	1.9	30

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Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence (J cm^{-2})	Measured Flu- ence (J cm^{-2})	Ablation Time (s)
2022-02-01	glass	43	5	3.5	3.6	30
2022-04-01	glass	43	5	3.5	3.4	30
2022-04-21	apatite	30	5	3.5	3.4	30
2022-04-21	zircon	30	5	2	2	30
2022-04-21	glass	43	5	3.5	3.4	30
2022-05-31	glass	43	5	5	5.2	30
2022-05-31	monazite	13	5	2	1.9	30
2022-05-31	xenotime	13	5	2	1.9	30
2022-06-20	zircon	30	5	2	2.1	30
2022-06-20	glass	43	5	3.5	3.6	30
2022-06-20	xenotime	13	5	2	2.1	30
2022-06-29	glass	43	5	3.5	3.4	30
2022-06-29	xenotime	13	5	2	1.9	30
2022-07-08	glass	43	5	3.5	3.6	40
2022-07-08	rutile	43	5	5	5.2	40
2022-08-30	monazite	20	5	2	2	30
2022-08-30	zircon	20	5	2	2	30
2022-10-10	glass	43	5	3.5	3.4	30
2022-10-10	zircon	30	5	2	1.9	30
2022-10-11	glass	30	5	3.5	3.4	30
2022-10-11	zircon	43	5	2	1.9	30
2022-12-09	glass	43	5	3.5	3.5	40
2022-12-09	rutile	43	5	5	4.9	40
2023-02-20	apatite	30	5	3.5	3.5	30
2023-02-20	glass	30	5	3.5	3.5	30
2023-02-20	zircon	30	5	2	1.9	30
2023-03-22	monazite	13	5	2	1.8	30
2023-03-22	xenotime	13	5	2	1.8	30

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Session Date	Material	Spot Diameter (μm)	Repetition Rate (Hz)	Nominal Fluence (J cm^{-2})	Measured Flu- ence (J cm^{-2})	Ablation Time (s)
2023-04-21	glass	43	5	3.5	3.5	30
2023-04-21	apatite	43	5	3.5	3.5	30
2023-04-21	zircon	30	5	2	2	30
2023-05-09	glass	43	5	3.5	3.5	40
2023-05-23	glass	43	5	3.5	3.6	40
2023-05-29	glass	43	5	3.5	3.4	30
2023-05-29	monazite	13	5	2	2.1	30
2023-06-30	glass	30	5	3.5	3.4	30
2023-06-30	apatite	30	5	3.5	3.4	30
2023-07-11	glass	43	5	3.5	3.5	40
2023-07-11	zircon	20	5	2	1.9	40
2023-11-24	apatite	30	5	3.5	3.5	30
2023-11-24	baddeleyite	30	5	2	2	30
2023-11-24	monazite	20	5	2	2	30
2023-11-24	rutile	43	5	5	4.9	30
2023-11-24	titanite	43	5	5	4.9	30
2023-11-24	xenotime	20	5	2	2	30
2023-11-24	zircon	30	5	2	2	30
2023-11-24	glass	30	5	3.5	3.5	30
2024-04-29	apatite	30	5	3.5	3.5	40
2024-04-29	baddeleyite	30	5	2	2.1	40
2024-04-29	cassiterite	43	5	5	5.1	40
2024-04-29	monazite	20	5	2	2.1	40
2024-04-29	rutile	43	5	5	5.1	40
2024-04-29	titanite	43	5	5	5.1	40
2024-04-29	xenotime	20	5	2	2.1	40
2024-04-29	zircon	30	5	2	2.1	40
2024-04-29	glass	30	5	3.5	3.5	40